Real-Time Seam Path Detection for Decorative Stitch Adjustment on Fabrics Using Image Processing

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Project Objective

Develop an intelligent system that detects the seam path on fabric in real time using ***image processing***, ***optimal control***, and ***dynamic programming***.

The system automatically adjusts decorative stitch positions according to existing swing line, ***eliminating manual alignment*** and improving precision, uniformity, and speed.

Then, Data exchange with Control Unit by ***CAN protocol***.

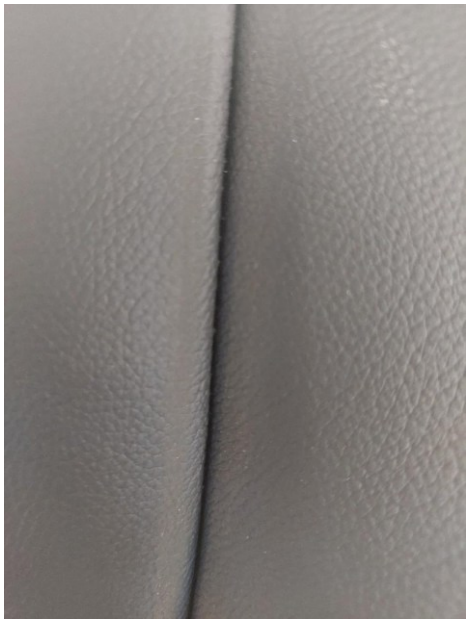
Sewed seam (before decoration stitching)

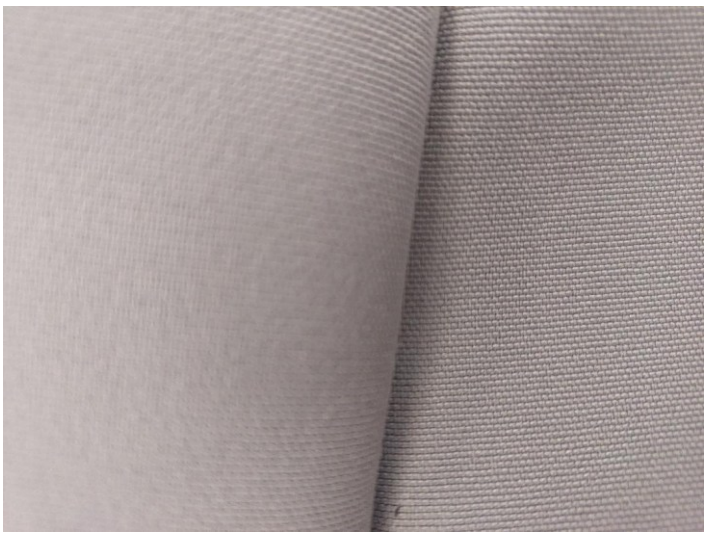
After decoration stitching (from Internet)

Fabric Patterns and textures

The system is designed to operate on a wide variety of fabrics with different materials, patterns, and textures.

Example samples are shown below.





Hardware Components

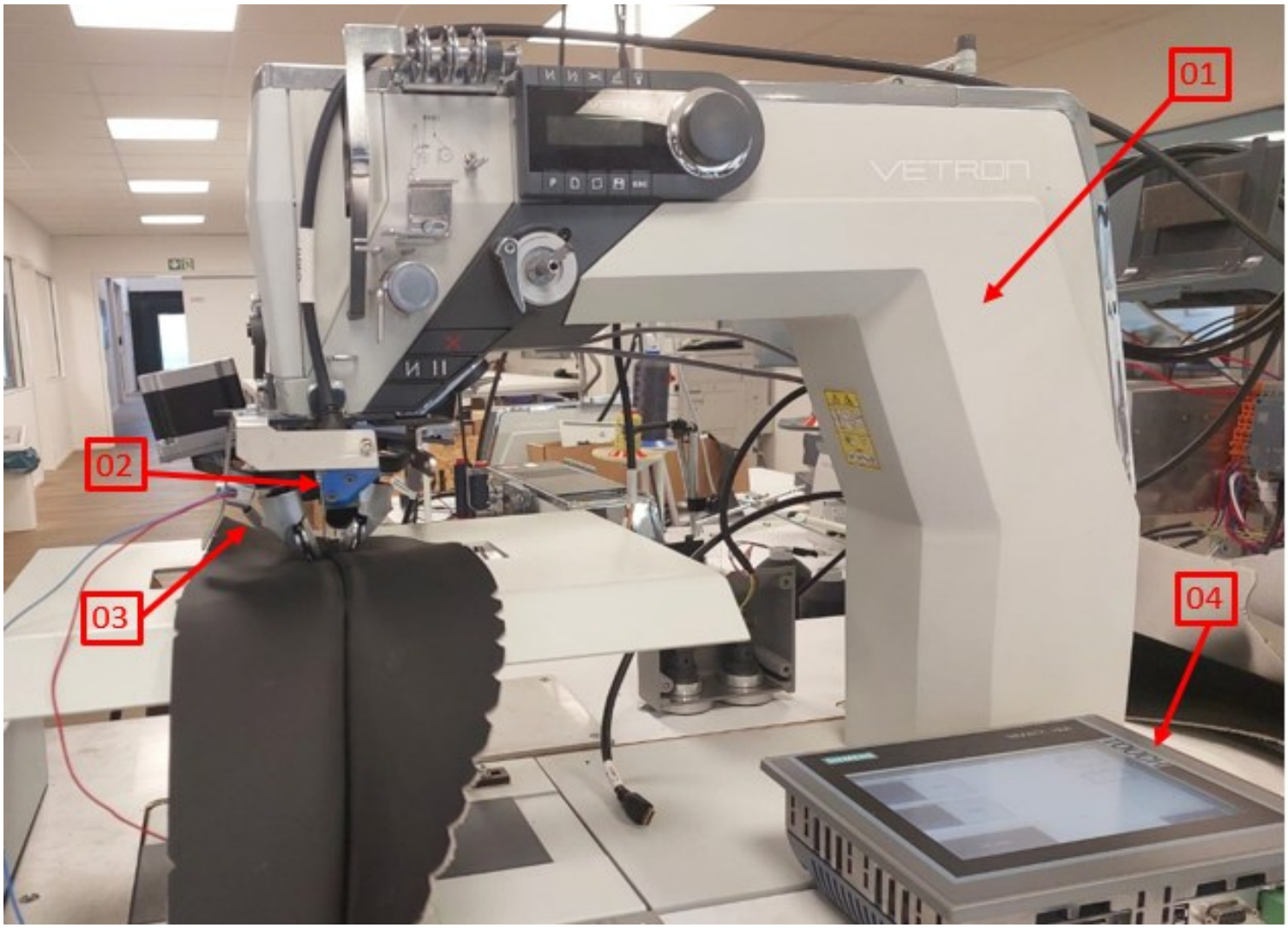
1. *Industrial Sewing Machine* – performs the stitching operation.

2. *XIMEA Industrial Camera* – captures high-resolution fabric images.

3. *Controlled Lighting Source* – provides uniform illumination.

4. Motion Controllers – manage fabric movement according to the detected path. (outside current project scope)

5. Processor/PC – executes real-time image processing algorithms.



Control Concepts

The algorithm applies **optimal control** principles using **dynamic programming** to locate the **lowest-cost vertical seam** in an image.

Each pixel's intensity represents a cost value, forming a **cumulative cost matrix** computed row by row.

A small penalty term encourages smoother, less zigzag seam paths.

The **optimal path** is obtained by backtracking from the bottom to the top of the matrix.

Bidirectional evaluation (left-to-right and right-to-left) enhances robustness.

Dynamic programming finds the lowest-cost vertical seam using a cumulative cost matrix C:

$$C(r,c)=E(r,c)+\min(C(r-1,c-1)+P,"\ "\ C(r-1,c),"\ "\ C(r-1,c+1)+P)$$

Wher:

- E(r,c): pixel cost (intensity)
- P: diagonal penalty

The Optimal path is backtracked from the last row

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```python
import sys
import os
import time
from datetime import datetime
import cv2
import numpy as np
from numba import njit

# --------------------
# XIMEA setup
# --------------------
sys.path.append(r"C:/Users/Sara.Bavi/Desktop/Sara/Ximea_MP9PU/
import xiapi

SAVE_DIR = r"C:\Users\Sara.Bavi\Desktop\Sara\Ximea_MP9PU\MY py
OUTPUT_DIR = r"C:\Users\Sara.Bavi\Desktop\Sara\Ximea_MP9PU\MY
os.makedirs(SAVE_DIR, exist_ok=True)
os.makedirs(OUTPUT_DIR, exist_ok=True)

# --------------------
# Config
# --------------------
GAUSS_KSIZE = 9
DIAG_PENALTY = 0.05
POLY_DEGREES = (1, 2, 3)
CLAHE_CLIP = 2.0
CLAHE_TILE = 8
PIXEL_SIZE_MM = 0.0022

# --------------------
# Seam finder
# --------------------
@njit
def find_vertical_dark_seam_numba(gray, diag_penalty=0.05):
    h, w = gray.shape
```

Image Processing Method

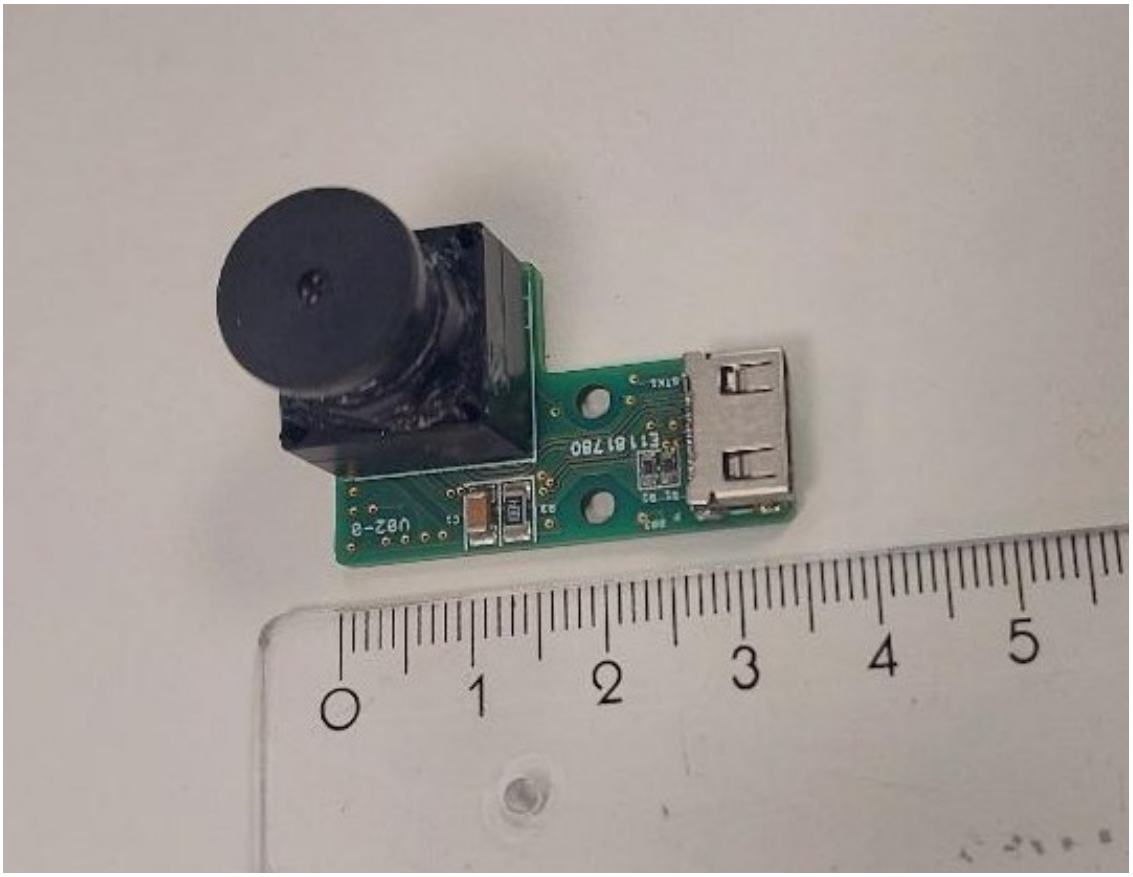
1. Image captured by the XIMEA camera in **MONO8** mode.

2. **CLAHE** (Contrast Limited Adaptive Histogram Equalization) enhances local contrast.

3. **Gaussian Blur** reduces small image noise.

4. **Gabor Filter** extracts texture and repetitive stitch patterns.

Result: a high-contrast, smooth image highlighting the seam path.



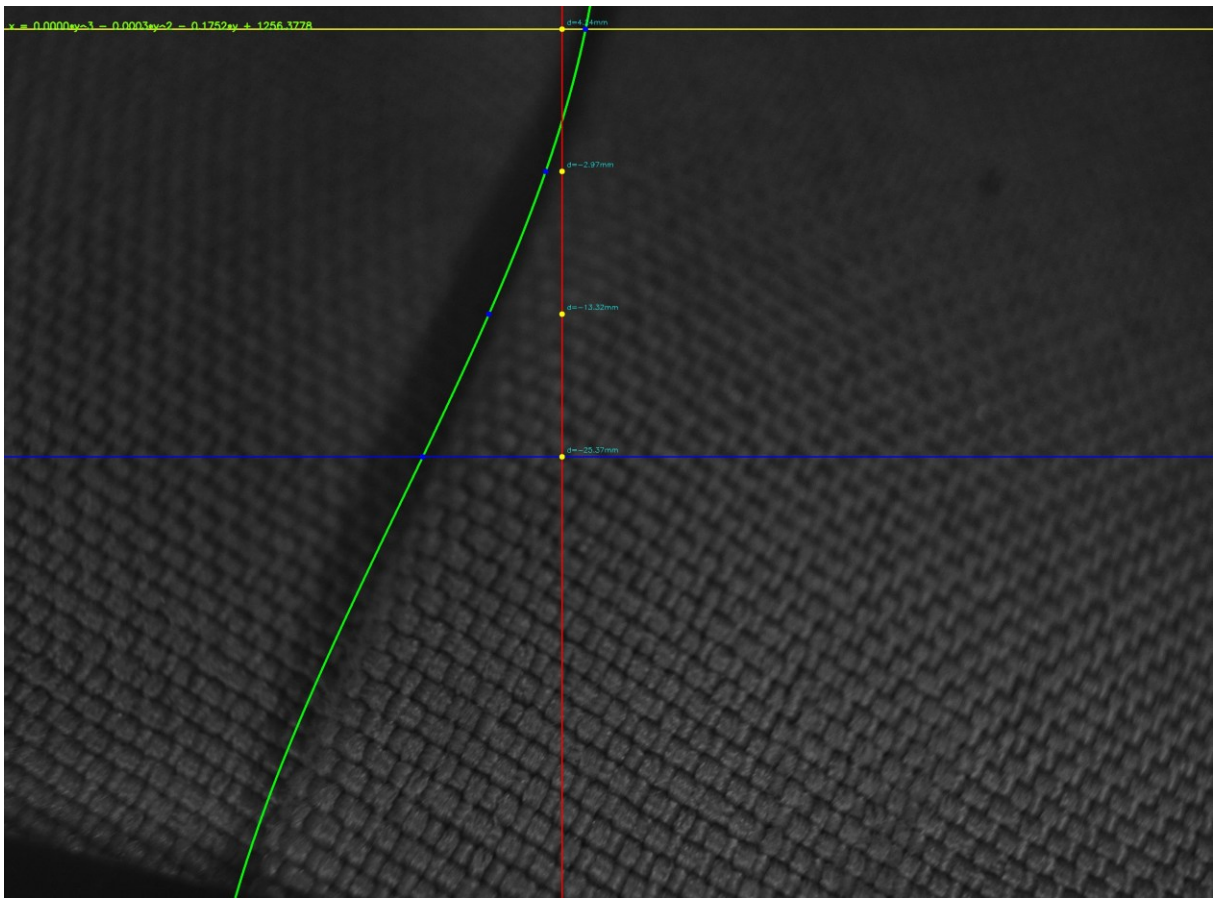
XIMEA Camera Mono-colour

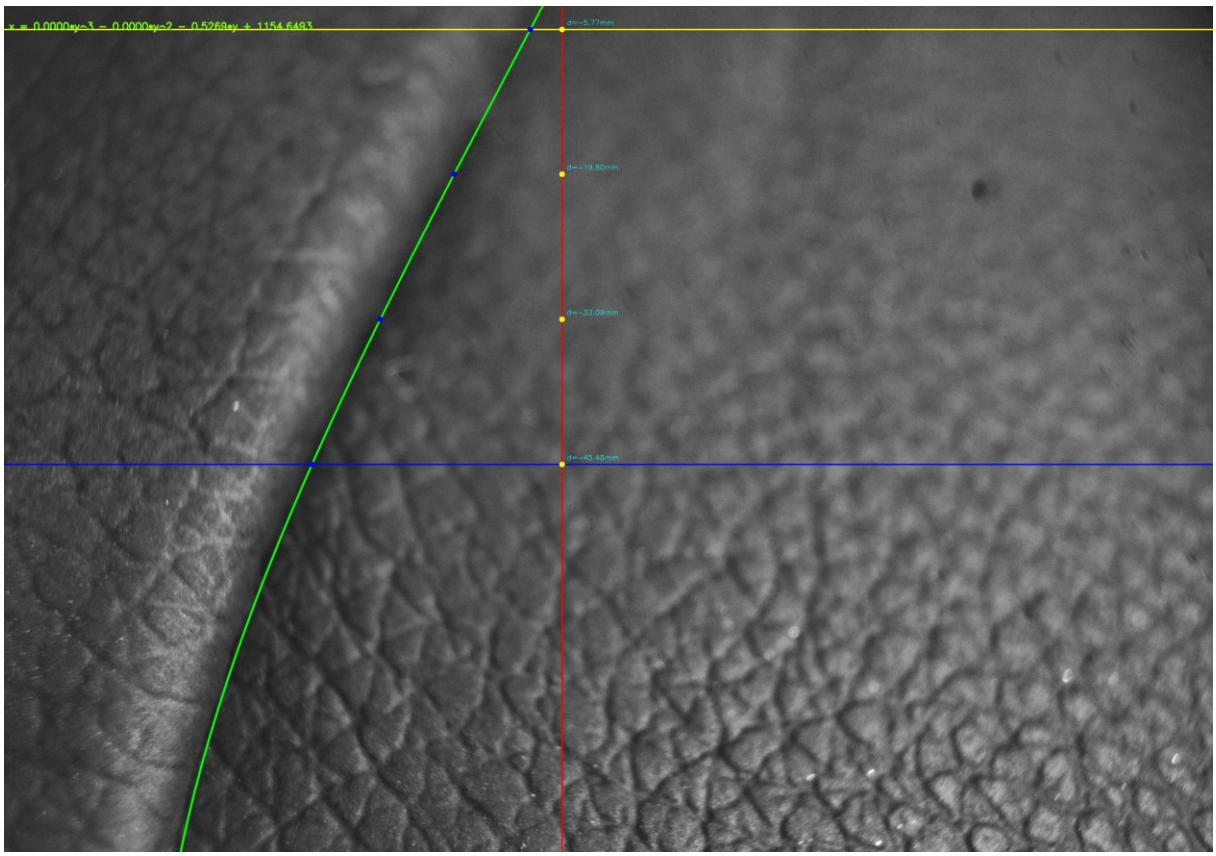
Seam Path Detection and Fitting

- Dark seam paths are detected using an ***optimized algorithm*** (find_vertical_dark_seam_with lowest cost map).
- The method runs bi-directionally (left-to-right and right-to-left) to produce a higher accuracy.
- RANSAC removes outlier points, followed by **polynomial fitting** (1st–3rd order) using polyfit for smooth curve estimation.
- **Poly-fit** applies also for smooth and continuous control.
- The **best-fitting curve** is selected based on minimum mean square error (**MSE**).

System Calibration, Path Control and Visualization

- The detected seam path is displayed as a green curve overlaid on the image.
- A red reference midline indicates the desired stitch trajectory.
- Deviation at four defined points is measured and converted to millimeters (0.0022 mm/pixel).
- The control system adjusts the needle or fabric to minimize deviation and ensure precise alignment.


x = 0.0000xy^3 − 0.0003xy^2 − 0.1752xy + 1256.3778
d=4.14mm
d=−2.97mm
d=−13.32mm
d=−25.37mm


x = 0.0000xy^3 − 0.0000xy^2 − 0.5268xy + 1154.5493
d=−5.77mm
d=−19.60mm
d=−33.09mm
d=−45.48mm

Challenges and Solutions

During system development, several practical and computational challenges were encountered, mainly related to lighting variations, fabric diversity, vibrations, and processing limitations.

The table below summarizes the main issues and the corresponding solutions implemented to improve detection accuracy and system stability.

Challenge	Solution
Lighting noise	Mitigated using **CLAHE** and **controlled illumination** to achieve consistent brightness and contrast.
Noise from different fabric patterns and textures	Reduced through **CLAHE enhancement** and **uniform lighting control**.
Fabric wrinkles during wheel motion	Corrected using **bidirectional detection** and **polynomial curve fitting** for path smoothing.
Vibrations from needle or fabric movement	Minimized by applying **frame-based temporal averaging** to stabilize image sequences.
High processing load and limited computational capacity	Optimized by using a **monochrome camera**, reducing image data and processing demand.
Missed seam detection on curved paths	Improved through **algorithm updates** enabling **dynamic path detection** and tracking.
Detection difficulty due to similarity between fabric pattern and real seam	Enhanced by integrating **RANSAC** and **MSE-based evaluation** into the detection algorithm.

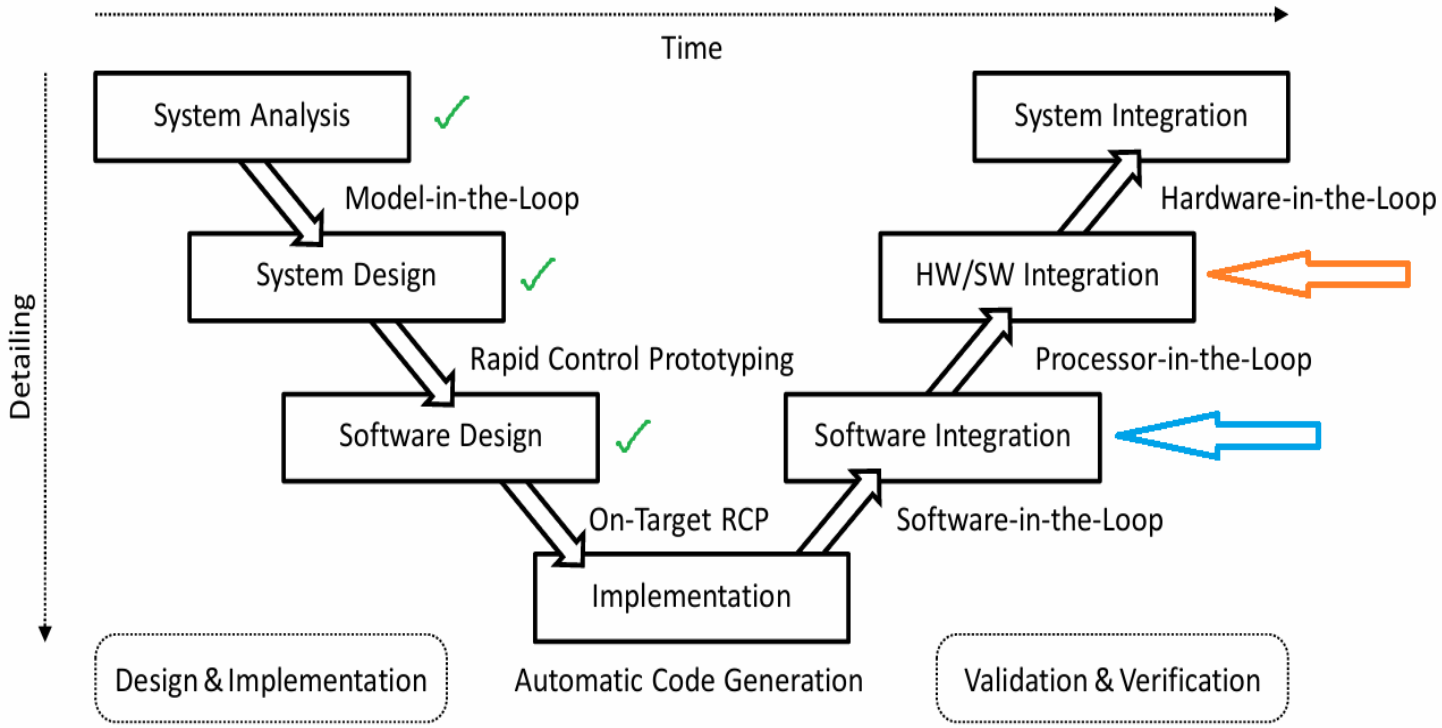
Expected Results and Applications

- Obtain accurate real-time seam path detection and control.
- Automated decorative stitching with high precision and uniformity.
- Increased production speed and product consistency.
- Applicable in textiles, upholstery, and automotive seat production.

Future Works

- Execute real-time stitching tests with integrated closed-loop control.
- Implement noise compensation through further filtering.
- Optimize algorithm for safe, reliability, and real-time efficiency in industrial environment.

- **V-Model**

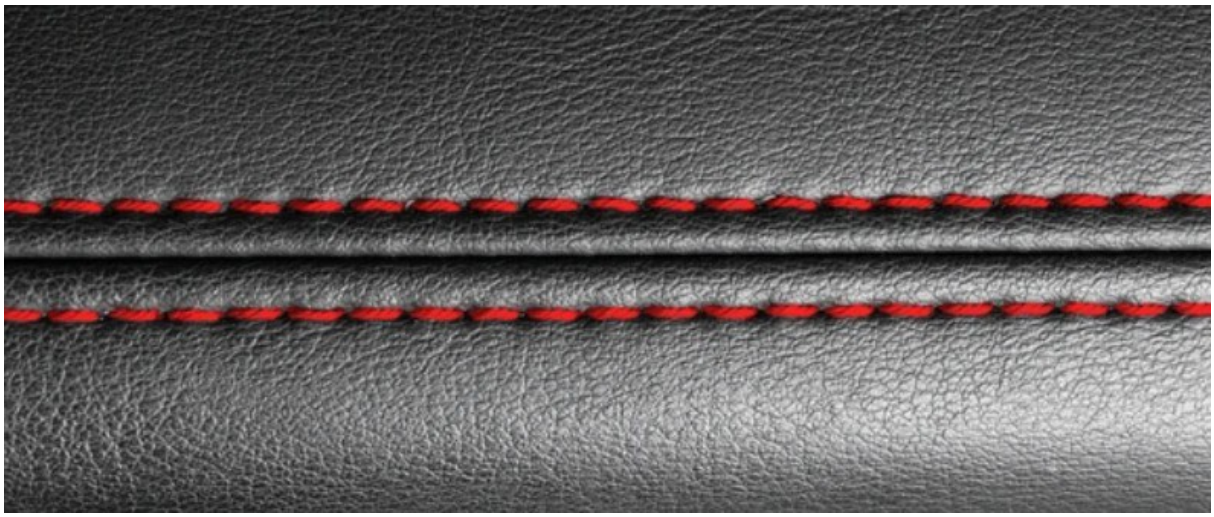


Recode the project in Linux: ubuntu

HW selection: NVIDIA or Rasberry Pi

Conclusion

The developed system provides intelligent, real-time seam path detection and automatic stitch adjustment, significantly enhancing **precision, consistency, and efficiency** in decorative sewing operations. This innovation **reduces manual alignment**, **minimizes human error**, and **substantially improves production quality and speed** in modern textile manufacturing.



The End